

8-Week Kids Coding, Basic Mathematics & AI Program Syllabus

Week 1 – Introduction to Programming & Arithmetic

- What is coding?
- Understanding algorithms
- Introduction to Scratch
- Basic Arithmetic: Addition and Subtraction
- Project: Animated greeting

Week 2 – Motion, Events, Loops & Arithmetic

- Moving sprites and keyboard controls
- Repeat and forever loops
- Basic Arithmetic: Multiplication and Division
- Project: Simple catching game

Week 3 – Variables, Conditions & Number Operations

- Variables and scores
- If/Else logic
- Comparing numbers (greater than, less than, equal to)
- Project: Quiz or maze game

Week 4 – Scratch Project & Math Challenge

- Build a game, story, or quiz
- Apply arithmetic scoring systems
- Presentation Day

Week 5 – Python Basics & Mathematics

- Introduction to Python
- print(), input(), variables
- Arithmetic operators (+, -, *, /)
- Project: Calculator and quiz game

Week 6 – Loops, Functions & Math Practice

- for loops and while loops
- Functions
- Multiplication tables and number patterns
- Project: Number guessing game

Week 7 – Introduction to AI & Responsible AI Use

- What is Artificial Intelligence?
- How AI learns from data
- Responsible use of AI
- Privacy and safety online
- AI strengths and limitations
- Activity: Improve a Python program using AI

Week 8 – AI Project Week & Digital Citizenship

- AI image classifier, AI-enhanced quiz, or chatbot
- Ethical use of AI
- Checking AI-generated information
- Final presentation
- Certificate award